Advanced Topics in Automation and Control Engineering

Aerial Robotics Laboratory

Self presentation

Dr Marco Tognon
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RAINBOW Team

Inria cnrs IRISA Université de Rennes

Background

POLITECNICO DI MILANO

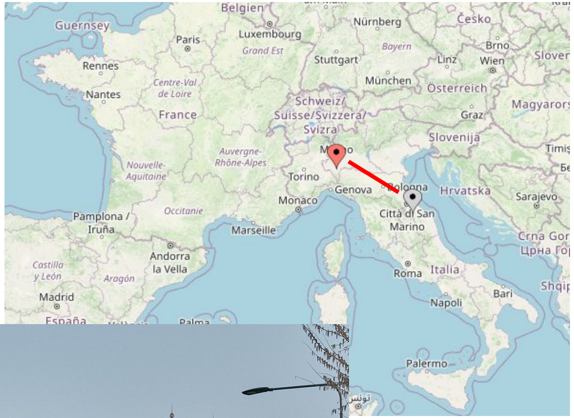
Milan (Italy)

In 2012

Bachelor in Automation and Control Engineering

In 2016

Master in Automation and Control Engineering



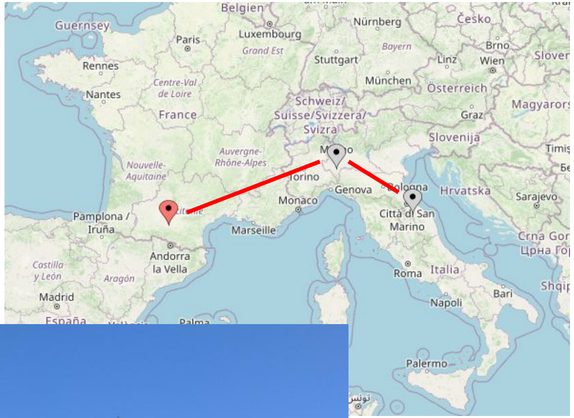
Background

LAAS - CNRS
Toulouse (France)

In 2019
6-month Master Internship

In the early 2020
CDD Engineer

In the late 2020
PhD Thesis
(Université Paul Sabatier III)



Background

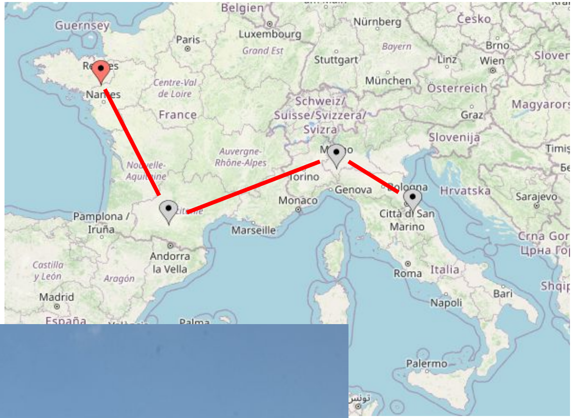
IRISA CNRS-INRIA

Rennes (France)

Since December 2023

CNRS Research Engineer

RAINBOW / SED



ROBSTAR

VISION ROBOTICS

Gantry robot
+ RGB-D camera

1992 – Retrofit 2008

AERIAL ROBOTICS

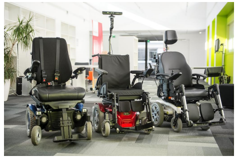
2015

2021

2024

+ Quadrotors
+ Fully actuated hexarotors
+ 2 Indoor flying arenas
+ 2 Motion capture systems

MOBILE ROBOTICS



2012

2018

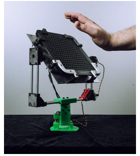
2025

+ 5 Wheelchairs
+ 1 Wheelchair simulator
+ 10 DJI RoboMaster

INTERACTIVES DEVICES & SYSTEMS

2016

2018



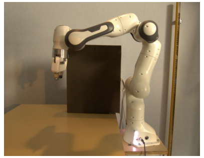
2018

2025

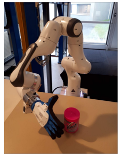
+ 1 Virtuose 6D
+ 1 Omega 6
+1 Ultrahaptics +2 Desktop 6D
+ VR & AR devices

MANIPULATION ROBOTICS

2009 - 2012



2017



2018

2018

2019

2025

2025

+ 2 Adept Viper
+ 3 Franka
+ 2 robotic hands
+ UR5

2025

TRIAGo
Multi-sensor
Multi-arm
Mobile
Manipulator

RAINBOW Team

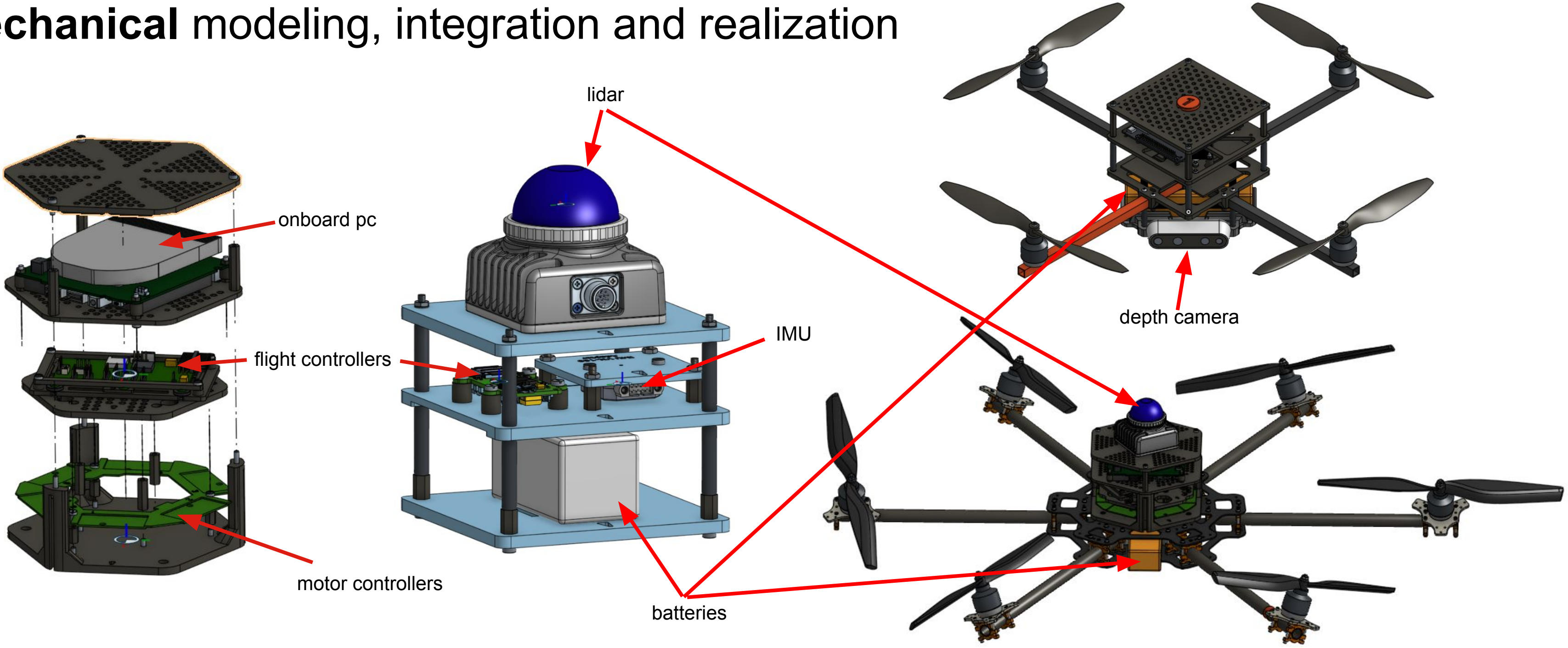
IRISA
Université de Rennes

Activities

Responsible for the **Aerial Robotic Platform**

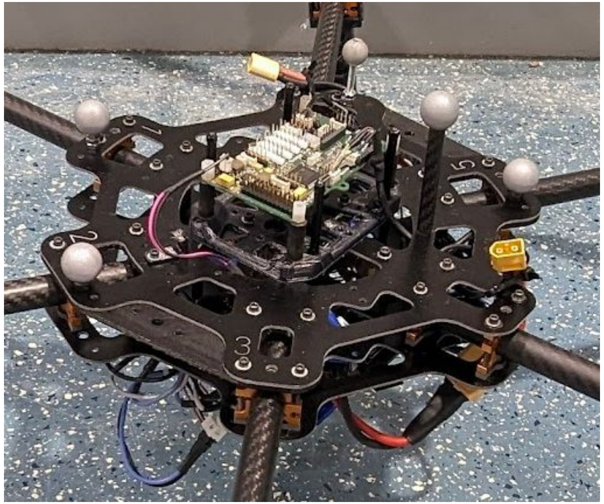
Activities

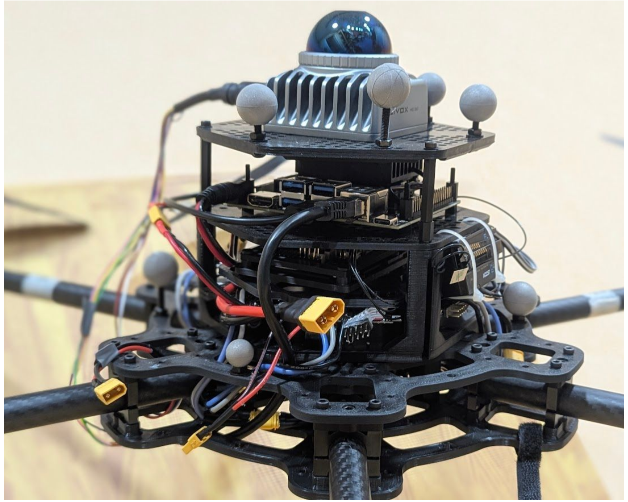
Mechanical modeling, integration and realization

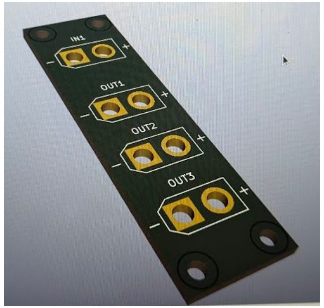


Activities

Electronic prototyping and integration







Activities

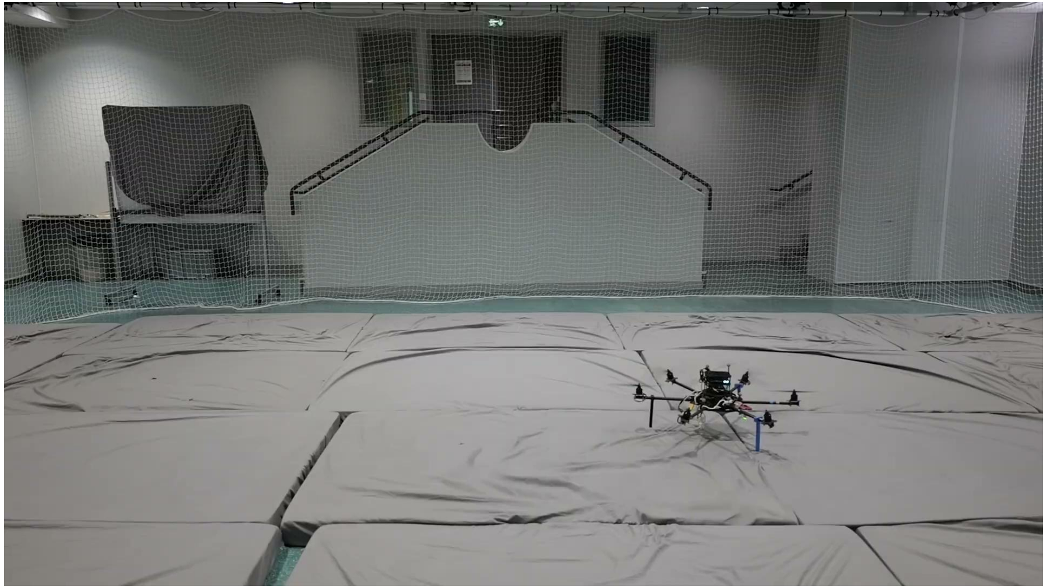
Software developer and maintainer (but not the author!)

telekyb3

- Open-source collection of software (and hardware) for multi-rotor unmanned aerial vehicles
- Used (mainly) in three major research laboratories: IRISA, LAAS, and the University of Twente
- Main applications: Control and navigation of one or more robots, physical interaction, and aerial manipulation

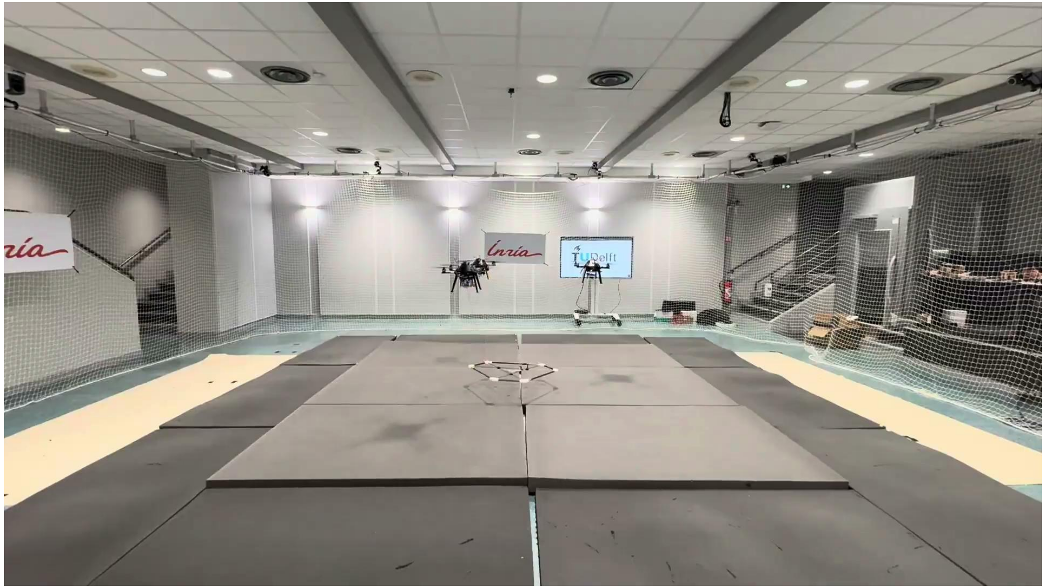
Experimental activities

TiltHex – On-board numerical optimization for agile trajectory maneuvering



Experimental activities

Flycrane – Agile aerial cooperative transportation and manipulation



Experimental activities

TiltODRI – Whole-body optimal trajectory generation